

Chen Wei Chang

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EDUCATION

M.S. National Yang Ming Chiao Tung University
Institute of Artificial Intelligence Innovation (GPA 4/4.3)

Hsinchu, Taiwan

Jul. 2023 - Aug. 2025

B.S. National Taipei University of Technology
Department of Electronics Engineering (GPA 3.91/4.0)

Taipei, Taiwan

Sep. 2019 - June. 2023

WORK EXPERIENCE

Full-Stack Engineer

Flask, React.js, MySQL, Linux

NYCU Mobile and Pervasive Computing Laboratory, Hsinchu, Taiwan

Sep. 2023 - Aug. 2025

- Developed multiple analysis features on CoachAI website, including scoring position replay, smash-to-net speed statistics, and a doubles website extension.
- Built a backend management system for account/group management, dataset authorization, and access control.
- Maintained the website on Linux servers.

Full-Stack Engineer

React.js, SQL Server, ASP.NET MVC

Mores Tech, Taipei, Taiwan

Feb. 2023 - Jun. 2023

- Developed a feature to generate various OpenOffice format reports for reference and download through LibreOffice integration.
- Developed a message and Q&A feature allowing users to report project updates to different departments through the system and track the progress of the projects.

PUBLICATIONS

- **LongE2V: Long-Horizon Event-based Video Reconstruction, Prediction, and Frame Interpolation with Video Diffusion Models** SIGGRAPH 2026
Cheng-De Fan, Chun-Wei Tuan Mu, Chen-Wei Chang, Chin-Yang Lin, Kun-Ru Wu, Yu-Chee Tseng, Yu-Lun Liu [Link](#)
LongE2V unifies event-based video reconstruction, prediction, and interpolation via pre-trained diffusion priors, curbing temporal drift in long sequences with Autoregressive Unrolling and Adaptive Context Switching.
- **GCC: Generative Color Constancy via Diffusing a Color Checker** CVPR 2025
Chen-Wei Chang, Cheng-De Fan, Chia-Che Chang, Yi-Chen Lo, Yu-Chee Tseng, Jiun-Long Huang, Yu-Lun Liu [Link](#)
A novel approach that leverages pretrained diffusion-based inpainting models to generate color checkers within images, enabling accurate illumination estimation for auto white balance across different camera sensors.
- **SpectroMotion: Dynamic 3D Reconstruction of Specular Scenes** CVPR 2025
Cheng-De Fan, Chen-Wei Chang, Yi-Ruei Liu, Jie-Ying Lee, Jiun-Long Huang, Yu-Chee Tseng, Yu-Lun Liu [Link](#)
A novel approach for dynamic specular 3D scene synthesis combining 3D Gaussian Splatting with physically-based rendering and deformation fields.
- **IllusionMaker** Under review at SIGGRAPH Asia 2025
Chun-Wei Tuan Mu, Zheng-Hui Huang, Chen-Wei Chang, Yung-Yu Chuang, Jia-Bin Huang, Yu-Lun Liu [Link](#)
We propose a diffusion-based model for spatially varying defocus blur editing that enables precise depth-of-field control and demonstrates applications in perspective illusion videos and DSLR-like bokeh simulation.

PROJECTS

- **Golf Ball Trajectory Estimation System**, Developed a computer vision system using high-speed cameras to track golf balls, calculate flight parameters, and visualize predicted flight trajectories on a web-based platform. [Link](#)
- **RESTful Web Game Platform**, Designed and implemented an interactive web gaming platform with user authentication, classic arcade games, and competitive leaderboards using HTML, CSS, JavaScript, Express.js, and SQLite. [Link](#)

EXTRACURRICULAR ACTIVITIES

- **Event Coordinator** in Student Association for the Department of Electrical Engineering, NTUT - Sep 2020 - Sep 2021

SELECTED HONORS AND AWARDS

- Master's Thesis Award, Taiwanese Association for Artificial Intelligence Dec. 2025
- Academic Achievement Award, Department of Electronics Engineering, NTUT May. 2020
- Academic Achievement Award, Department of Electronics Engineering, NTUT Nov. 2020

TECHNICAL SKILLS

Programming Languages: Python, C++, JavaScript(React.js), SQL(MySQL, SQL server), Verilog

Libraries and Tools: PyTorch, OpenCV, Git, Linux